

§1.2 Vector Spaces and Subspaces

Motivation: In §1.1, we saw the properties of $+$ and $\cdot$ as operations on $\mathbb{R}^n$ (Prop 1.6 (VS1) – (VS8)).

As said before, want

- Vector: ^{elts}elements that can be added & scalar-multiplied
- Vector Space: suitable set of vectors

Q: What else behaves like $\mathbb{R}^n$?

A: Functions ^{fns} $f, g: \mathbb{R} \rightarrow \mathbb{R}$ can be added & scalar-multiplied:

$$\begin{array}{ccc} \underline{f+g}: \mathbb{R} \rightarrow \mathbb{R}, & & \underline{c \cdot f}: \mathbb{R} \rightarrow \mathbb{R} \\ \uparrow & x \mapsto f(x) + g(x) & \uparrow \\ & & x \mapsto c f(x) \end{array}$$

both are fns! They "look like vectors".

Example

Ex 1.8: Let X be a set. Define

$\mathcal{F}(X, \mathbb{R}) :=$ set of all fns from X to $\mathbb{R}$.

For instance, $h: \mathbb{R} \rightarrow \mathbb{R}$ is in $\mathcal{F}(\mathbb{R}, \mathbb{R})$.
 $x \mapsto |x|$

Def $+$: $\mathcal{F}(X, \mathbb{R}) \times \mathcal{F}(X, \mathbb{R}) \rightarrow \mathcal{F}(X, \mathbb{R})$

$(f, g) \mapsto \left(\begin{array}{l} f+g: X \rightarrow \mathbb{R} \\ x \mapsto f(x) + g(x) \end{array} \right)$
^{scalar-mult} sends left to right

$\cdot: \mathbb{R} \times \mathcal{F}(X, \mathbb{R}) \rightarrow \mathcal{F}(X, \mathbb{R})$

$(c, f) \mapsto \left(\begin{array}{l} cf: X \rightarrow \mathbb{R} \\ x \mapsto cf(x) \end{array} \right)$

Claim: $\mathcal{F}(X, \mathbb{R})$ w/ $+$ & $\cdot$ satisfy (VS1) – (VS8)

from Prop 1.6 when replacing $\mathbb{R}^n$ by $\mathcal{F}(X, \mathbb{R})$.

(And in (VS3), $0_{\mathcal{F}(X, \mathbb{R})} = 0: X \rightarrow \mathbb{R}$,
 $x \mapsto 0$
in (VS4), $w = (-1) \cdot v: X \rightarrow \mathbb{R}$
 $x \mapsto (-1)v(x)$) } Exercise

Proof

Pf Idea: $+$ & $\cdot$ are defined via outputs of fns,
which live in $\mathbb{R}$. And $\mathbb{R}$ satisfies (VS1) – (VS8),
so $\mathcal{F}(X, \mathbb{R})$ satisfies (VS1) – (VS8) as well.

Actual Pf of (VS1) (rest are Exercises):

For $\forall$ every $f, g \in \mathcal{F}(X, \mathbb{R})$, $\forall x \in X$,

$(f+g)(x) = f(x) + g(x) = g(x) + f(x) = (g+f)(x)$, so

$f+g = g+f$. □

Conclusion: (VS1) – (VS8) from Prop 1.6 (in §1.1)
are rules that determine when $+$ & $\cdot$ behave like
 $+$ & $\cdot$ from $\mathbb{R}^n$.

Definition

Def 1.9: A vector space V (over $\mathbb{R}$) is a set

w/ addition $+: V \times V \rightarrow V$ and scalar-mult

$\cdot: \mathbb{R} \times V \rightarrow V$ satisfying (VS1) – (VS8) are called
axioms of v.s.

$$(VS 1) \forall v, w \in V, v + w = w + v$$

$$(VS 2) \forall u, v, w \in V, (u + v) + w = u + (v + w)$$

$$(VS 3) \text{ There } \exists \text{ exists } \underline{0_V} \in V \text{ such } \text{s.t.} \text{ that } \forall v \in V, \\ v + 0_V = v = 0_V + v \text{ called zero-vector}$$

$$(VS 4) \forall v \in V, \exists \underline{w} \in V \text{ s.t. } v + w = 0_V \\ \text{called additive inverse of } v$$

$$(VS 5) \forall v \in V, 1 \cdot v = v$$

$$(VS 6) \forall a, b \in \mathbb{R} \text{ and } \forall v \in V, (ab) \cdot v = a \cdot (b \cdot v) \\ \text{with } (ab)v \text{ and } a(bv) \text{ indicated above}$$

$$(VS 7) \forall c \in \mathbb{R} \text{ and } \forall v, w \in V, c(v + w) = cv + cw$$

$$(VS 8) \forall a, b \in \mathbb{R} \text{ and } \forall v \in V, (a + b)v = av + bv$$

Elements ^{elts} of V are called vectors ^{vecs} (in V).

Remark

Rmk 1.10:

$$(a) \exists \text{ } \underline{\text{1-1 correspondence}} \text{ } \mathbb{R}^n \xleftrightarrow{\text{1-1 \& onto fun}} \mathcal{F}(\{1, \dots, n\}, \mathbb{R}) \\ (a_1, \dots, a_n) \mapsto \left(a: \{1, \dots, n\} \rightarrow \mathbb{R} \right. \\ \left. \quad \quad \quad \bar{i} \mapsto a_i \right) \\ (f(1), \dots, f(n)) \leftarrow (f: \{1, \dots, n\} \rightarrow \mathbb{R})$$

Pf: Exercise

□

(b) $\mathbb{R}^n$ is a v.s. by Prop 1.6. So $\mathbb{R}^n$ is a v.s. of column ^{col} vecs of size n (also a v.s. of n -tuples of ^{$\mathbb{R}$} real ^{$\#s$} numbers).

(c) $\mathcal{F}(X, \mathbb{R})$ is a v.s. of fcn's $X \rightarrow \mathbb{R}$ by Ex 1.8.

$\uparrow$ generalizes $\mathbb{R}^n$ by (a)

Note: NO NEED TO MEMORIZE Statement #s (including axiom #s), as they depend on which books/notes. Those #s are NOT important!

What's IMPORTANT is the CONTENT.

Ex 1.11: Let $\mathbb{R}^\infty := \mathcal{F}(\underbrace{\mathbb{Z}_{>0}}_{\{1,2,3,\dots\}}, \mathbb{R})$ be a v.s. (by Ex 1.8). Then just like Rmk 1.10(a), $\forall a \in \mathbb{R}^\infty$ can be thought of as a ^{seq} sequence $a(1), a(2), \dots$. Thus, seqs of $\mathbb{R}$ #s "are" vecs in $\mathbb{R}^\infty$.

Ex 1.12: A_n $\overset{m,n \in \mathbb{Z}_{>0}}{(m) \times (n)}$ ($\mathbb{R}$ -) matrix is a rectangular

array of $\mathbb{R}$ #s w/ m rows and n cols:

$$A = \begin{bmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & & \vdots \\ \text{\textcolor{violet}{\textit{i}th row}} \vdots & \text{\textcolor{teal}{\textit{A}_{ij}}} & \vdots \\ A_{m1} & \cdots & A_{mn} \end{bmatrix} \quad \begin{array}{l} \text{\textcolor{teal}{\textit{A}_{ij}}} = \text{\textcolor{teal}{(A)_{ij}}} \text{ means the} \\ \text{\textcolor{violet}{(i)}} \text{\textcolor{blue}{(j)}} \text{-entry of } A. \end{array}$$

The set $M_{m \times n}(\mathbb{R})$ of all $m \times n$ matrices w/

$$\diamond A+B \text{ def by } (A+B)_{ij} := A_{ij} + B_{ij},$$

◇ cA def by $(cA)_{ij} := c(A_{ij})$,

is a v.s.

Pf: Exercise (Similar to Pf of Ex 1.8.) $\square$

Rmk 1.13:

(a) $\mathbb{R}^n = M_{n \times 1}(\mathbb{R})$ by def.

(b) A row vec of size n is $[a_1 \cdots a_n] \in M_{1 \times n}(\mathbb{R})$.

For example, $[1 \ 3 \ 5] \in M_{1 \times 3}(\mathbb{R})$ v.s. of row
vecs

$$\begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} = (1, 3, 5) \in \mathbb{R}^3 = M_{3 \times 1}(\mathbb{R})$$

(c) $M_n(\mathbb{R}) := M_{n \times n}(\mathbb{R})$ v.s. of square matrices
of size n .

Ex 1.14: Consider $\mathbb{R}^2$ w/ usual scalar-mult.

and new addition $+' : \mathbb{R}^2 \times \mathbb{R}^2 \longrightarrow \mathbb{R}^2$.

$$(a, b), (c, d) \mapsto (a+c, 0)$$

Then (VS3) fails for $v = (1, 1)$, because ^{b/c}

$\forall (c, d) \in \mathbb{R}^2, (1, 1) + (c, d) = (1+c, 0) \neq (1, 1)$, so

^{not $\exists$} $0_{\mathbb{R}^2}$ for $(1, 1) \in \mathbb{R}^2$ w/ $+'$ & .

Properties of V.S.

Motivation: For $\mathbb{R}^n$ & $\mathcal{F}(X, \mathbb{R})$, 0 from (VS3) and additive inverse from (VS4) are unique.

Also, $a+c = b+c \Rightarrow a=c$ on those two Exs.

Q: True for general v.s.?

$\Downarrow$ Yes!